

Exercise of Supervised Learning: Information Theory Part 2

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Exercise 1: Entropy

A fair die is rolled at the same time as a fair coin is tossed. Let A be the number on the upper surface of the dice and let B describe the outcome of the coin toss, where

$$B = \begin{cases} 1, & \text{head,} \\ 0 & \text{tail.} \end{cases}$$

Two random variables X and Y are given by $X = A + B$ and $Y = A - B$, respectively.

(a) Calculate the entropies $H(X)$ and $H(Y)$, the conditional entropies $H(Y|X)$ and $H(X|Y)$, the joint entropy $H(X, Y)$ and the mutual information $I(X; Y)$.

Exercise 1 (a): Setup

Definitions

$$X = A + B, \quad Y = A - B$$

Elementary probabilities

$$P(A = a) = \frac{1}{6}, \quad a = 1, \dots, 6,$$

$$P(B = b) = \frac{1}{2}, \quad b = 0, 1,$$

$$P(A = a, B = b) = \frac{1}{12}.$$

Before $H(X)$ and $H(Y)$

Compute the joint distribution of X and Y by using the relation between (x, y) and (a, b) .

One-to-one joint mapping

$$(a, b) \mapsto (x, y) = (a + b, a - b)$$

$$(x, y) \mapsto (a, b) = \left(\frac{x + y}{2}, \frac{x - y}{2} \right)$$

Consequence

$$P(X = x, Y = y) = P(A = a, B = b) = \frac{1}{12}$$

for every feasible (x, y) .

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Exercise 1 (a): Joint Entropy

Joint entropy

$$\begin{aligned} H(X, Y) &= - \sum_{x,y} p_{X,Y}(x, y) \log_2 p_{X,Y}(x, y) \\ &= -12 \cdot \frac{1}{12} \log_2 \frac{1}{12} \\ &= 2 + \log_2 3 \end{aligned}$$

Key fact

$$(a, b) \leftrightarrow (x, y)$$

$$p(x, y) = p(a, b) = \frac{1}{12}.$$

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Key fact

$$(a, b) \leftrightarrow (x, y)$$

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Exercise 1(a): $H(X)$

- ▶ Many (a, b) map to one x .
e.g. $x=2$: $(2, 0), (1, 1)$.
- ▶ Each pair: $p_A(a) p_B(b) = \frac{1}{12}$.
- ▶ Group by $x=a+b$:

$$p_X(x) = \frac{1}{12} |\{(a, b) : a+b=x\}|.$$

- ▶ Plug into

$$H(X) = -\sum_x p_X(x) \log_2 p_X(x).$$

x	Events (a, b)	$p_X(x)$
1	$(1, 0)$	$1/12$
2	$(2, 0), (1, 1)$	$1/6$
3	$(3, 0), (2, 1)$	$1/6$
4	$(4, 0), (3, 1)$	$1/6$
5	$(5, 0), (4, 1)$	$1/6$
6	$(6, 0), (5, 1)$	$1/6$
7	$(6, 1)$	$1/12$

$$\begin{aligned} H(X) &= -2 \cdot \frac{1}{12} \log_2 \frac{1}{12} - 5 \cdot \frac{1}{6} \log_2 \frac{1}{6} \\ &= \frac{7}{6} + \log_2 3. \end{aligned}$$

Exercise 1(a): $H(Y)$

- ▶ Many (a, b) map to one y .
e.g. $y=1$: $(1, 0), (2, 1)$.
- ▶ Each pair: $p_A(a) p_B(b) = \frac{1}{12}$.
- ▶ Group by $y=a-b$:

$$p_Y(y) = \frac{1}{12} |\{(a, b) : a-b=y\}|.$$

- ▶ Plug into

$$H(Y) = -\sum_y p_Y(y) \log_2 p_Y(y).$$

y	Events (a, b)	$p_Y(y)$
0	$(1, 1)$	$1/12$
1	$(1, 0), (2, 1)$	$1/6$
2	$(2, 0), (3, 1)$	$1/6$
3	$(3, 0), (4, 1)$	$1/6$
4	$(4, 0), (5, 1)$	$1/6$
5	$(5, 0), (6, 1)$	$1/6$
6	$(6, 0)$	$1/12$

$$\begin{aligned} H(Y) &= -2 \cdot \frac{1}{12} \log_2 \frac{1}{12} - 5 \cdot \frac{1}{6} \log_2 \frac{1}{6} \\ &= \frac{7}{6} + \log_2 3. \end{aligned}$$

Exercise 1(a): $H(X|Y)$, $H(Y|X)$, $I(X; Y)$

Recall

- Chain rule:

$$\begin{aligned}H(X, Y) &= H(X) + H(Y|X) \\ &= H(Y) + H(X|Y)\end{aligned}$$

- Mutual information:

$$I(X; Y) = H(X) - H(X|Y)$$

Plug in $H(X, Y) = 2 + \log_2 3$ and
 $H(X) = H(Y) = \frac{7}{6} + \log_2 3$:

$$H(X|Y) = H(X, Y) - H(Y) = \frac{5}{6}$$

$$H(Y|X) = H(X, Y) - H(X) = \frac{5}{6}$$

$$I(X; Y) = H(X) - H(X|Y) = \frac{1}{3} + \log_2 3$$

Exercise 1 (b)

Question. For independent discrete random variables X and Y , show

$$I(X; X + Y) - I(Y; X + Y) = H(X) - H(Y).$$

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$$I(X; X + Y) - I(Y; X + Y) = H(X + Y) - H(X + Y|X) - H(X + Y) + H(X + Y|Y).$$

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Conditional entropy step

$$p(x + y|x) = p(y|x)$$

Given x , knowing $x + y$ is the same
as knowing y :

$$H(X + Y|X) = H(Y|X).$$

Similarly,

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Remaining calculation

$$H(X + Y|Y) - H(X + Y|X)$$

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$$\begin{aligned} &H(X + Y|Y) - H(X + Y|X) \\ &= H(X|Y) - H(Y|X) \end{aligned}$$

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Given x , knowing $x + y$ is the same
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Similarly,

$$H(X + Y|Y) = H(X|Y).$$

Remaining calculation

$$H(X + Y|Y) - H(X + Y|X)$$

$$= H(X|Y) - H(Y|X)$$

$$= H(X) - H(Y),$$

because X and Y are independent.

Exercise 2 (a)

Let X , Y and Z be three discrete random variables. The mutual information of X , Y and Z is defined as:

$$I(X; Y; Z) = \sum_x \sum_y \sum_z p(x, y, z) \log \left(\frac{p(x, y)p(x, z)p(y, z)}{p(x)p(y)p(z)p(x, y, z)} \right)$$

(a) Prove the lemma: $I(X; Y; Z) = I(X; Y) - I(X; Y|Z)$. Note that the conditional mutual information is defined as:

$$I(X; Y|Z) = \sum_x \sum_y \sum_z p(z)p(x, y|z) \log \frac{p(x, y|z)}{p(x|z)p(y|z)}$$

Hint: Starting from the right hand side of the equation in the lemma.

Exercise 2 (a): Proof of $I(X; Y; Z) = I(X; Y) - I(X; Y|Z)$

$$I(X; Y) - I(X; Y|Z) = \sum_x \sum_y p(x, y) \log \frac{p(x, y)}{p(x)p(y)} - \sum_z \sum_x \sum_y p(z)p(x, y|z) \log \frac{p(x, y|z)}{p(x|z)p(y|z)}$$

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$$\begin{aligned} I(X; Y) - I(X; Y|Z) &= \sum_x \sum_y p(x, y) \log \frac{p(x, y)}{p(x)p(y)} - \sum_z \sum_x \sum_y p(z)p(x, y|z) \log \frac{p(x, y|z)}{p(x|z)p(y|z)} \\ &= \sum_{x, y, z} p(x, y, z) \log \frac{p(x, y)}{p(x)p(y)} - \sum_{x, y, z} p(x, y, z) \log \frac{p(x, y|z)p(z)^2}{p(x|z)p(y|z)p(z)^2} \end{aligned}$$

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Exercise 2 (b)

Question. Prove

$$I(X; Y) = I(X; Y|Z) + I(Y; Z) - I(Y; Z|X).$$

Lemma

$$I(X; Y; Z) = I(X; Y) - I(X; Y|Z).$$

Computation. Apply the lemma twice:

$$I(X; Y|Z) = I(X; Y) - I(X; Y; Z), \quad I(Y; Z|X) = I(Y; Z) - I(X; Y; Z).$$

Then

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Then

$$\begin{aligned} & I(X; Y|Z) + I(Y; Z) - I(Y; Z|X) \\ &= (I(X; Y) - I(X; Y; Z)) + I(Y; Z) - (I(Y; Z) - I(X; Y; Z)) \end{aligned}$$

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Then

$$\begin{aligned} & I(X; Y|Z) + I(Y; Z) - I(Y; Z|X) \\ &= (I(X; Y) - I(X; Y; Z)) + I(Y; Z) - (I(Y; Z) - I(X; Y; Z)) \\ &= I(X; Y). \end{aligned}$$

Exercise 3: Smoothed Cross-Entropy Loss

Over-confidence: model is more confident in its prediction than the data warrants. **Label smoothing** (Szegedy et al., 2016) alleviates this and improves robustness.

Conventional cross-entropy minimizes the KL-divergence between the ground-truth distribution d (a delta: $d_k = 1$ for the true class) and the predicted distribution $\pi(\mathbf{x}|\boldsymbol{\theta})$.

Label smoothing smooths d : given β (e.g. $\beta = 0.1$), distribute probability mass β uniformly across the g classes and reduce the ground-truth mass:

$$\tilde{d}_k = \begin{cases} \beta/g & \text{for } d_k = 0, \\ 1 - \beta + \beta/g & \text{for } d_k = 1. \end{cases}$$

The smoothed cross-entropy is then $D_{KL}(\tilde{d} \parallel \pi(\mathbf{x}|\boldsymbol{\theta}))$.

(a) Derive the empirical risk when using the smoothed cross-entropy as loss function. *Hint: some terms can be merged into a constant and ignored during implementation.*

Exercise 3 (a): $D_{KL}(\tilde{d} \parallel \pi(\mathbf{x}|\theta))$

Start from the KL definition and expand:

$$\begin{aligned} R_{\text{emp}}(\theta) &= \frac{1}{n} \sum_{i=1}^n \sum_{k=1}^g \tilde{d}_k^{(i)} \log \frac{\tilde{d}_k^{(i)}}{\pi_k(\mathbf{x}^{(i)}|\theta)} \\ &= \frac{1}{n} \sum_{i=1}^n \sum_{k=1}^g \left(\underbrace{\tilde{d}_k^{(i)} \log \tilde{d}_k^{(i)}}_{\text{indep. of } \theta} - \tilde{d}_k^{(i)} \log \pi_k(\mathbf{x}^{(i)}|\theta) \right). \end{aligned}$$

Recall: $\tilde{d}_k$

$$\tilde{d}_k = \begin{cases} \beta/g, & d_k = 0 \\ 1 - \beta + \beta/g, & d_k = 1 \end{cases}$$

Drop the first term (constant w.r.t. θ) and plug in $\tilde{d}_k$ (with $y^{(i)}$ the true class) to get the practical form:

$$R_{\text{emp}}(\theta) = -\frac{1}{n} \sum_{i=1}^n \left[(1 - \beta) \log \pi_{y^{(i)}}(\mathbf{x}^{(i)}|\theta) + \frac{\beta}{g} \sum_{k=1}^g \log \pi_k(\mathbf{x}^{(i)}|\theta) \right] + \text{const.}$$

Exercise 3 (a): $D_{KL}(\tilde{d} \parallel \pi(\mathbf{x}|\boldsymbol{\theta}))$

Start from the KL definition and expand:

$$\begin{aligned} R_{\text{emp}}(\boldsymbol{\theta}) &= \frac{1}{n} \sum_{i=1}^n \sum_{k=1}^g \tilde{d}_k^{(i)} \log \frac{\tilde{d}_k^{(i)}}{\pi_k(\mathbf{x}^{(i)}|\boldsymbol{\theta})} \\ &= \frac{1}{n} \sum_{i=1}^n \sum_{k=1}^g \left(\underbrace{\tilde{d}_k^{(i)} \log \tilde{d}_k^{(i)}}_{\text{indep. of } \boldsymbol{\theta}} - \tilde{d}_k^{(i)} \log \pi_k(\mathbf{x}^{(i)}|\boldsymbol{\theta}) \right). \end{aligned}$$

Recall: $\tilde{d}_k$

$$\tilde{d}_k = \begin{cases} \beta/g, & d_k = 0 \\ 1 - \beta + \beta/g, & d_k = 1 \end{cases}$$

Drop the first term (constant w.r.t. $\boldsymbol{\theta}$) and plug in $\tilde{d}_k$ (with $y^{(i)}$ the true class) to get the practical form:

$$R_{\text{emp}}(\boldsymbol{\theta}) = -\frac{1}{n} \sum_{i=1}^n \left[(1 - \beta) \log \pi_{y^{(i)}}(\mathbf{x}^{(i)}|\boldsymbol{\theta}) + \frac{\beta}{g} \sum_{k=1}^g \log \pi_k(\mathbf{x}^{(i)}|\boldsymbol{\theta}) \right] + \text{const.}$$

Exercise 3 (b): Implementation (Python)

(b) Implement the smoothed cross-entropy. Vectorised numpy version:

```
import numpy as np
from numpy.typing import NDArray

def smoothed_ce_loss(label: NDArray,      # shape (n,), values in {1, ..., g}
                    pred: NDArray,        # shape (n, g), predicted probs
                    smoothing: float) -> float:
    assert 0.0 <= smoothing <= 1.0
    n, g = pred.shape
    assert label.shape == (n,)

    # Build smoothed label matrix: start uniform, set true class to 1 - b + b/g
    base = smoothing / g
    smoothed = np.full((n, g), base)
    smoothed[np.arange(n), label - 1] = 1.0 - smoothing + base

    # Per-sample cross-entropy, then mean over the batch
    return float(-np.sum(smoothed * np.log(pred), axis=1).mean())
```

Exercise 3 (b): Demo Runs

A “confident” toy model: 5 samples, 3 classes.

```
label = np.array([1, 2, 2, 3, 1])
pred  = np.array([[0.85, 0.10, 0.05],
                  [0.05, 0.90, 0.05],
                  [0.02, 0.95, 0.03],
                  [0.13, 0.02, 0.85],
                  [0.86, 0.04, 0.10]])

smoothed_ce_loss(label, pred, smoothing=0.0)  # plain CE      -> 0.1265
smoothed_ce_loss(label, pred, smoothing=0.2)  # label-smoothed -> 0.5121
```

Takeaway

- ▶ Loss jumps $\sim 4\times$ ($0.13 \rightarrow 0.51$) with $\beta = 0.2$.
- ▶ Target is no longer a delta $\Rightarrow$ a confident peak on the true class is now *penalised*.

Summary — Connecting the exercise to the lecture

	What we derived	Lecture concept made concrete
1 (a)	$H(X), H(Y X), H(X, Y), I(X; Y)$ from the joint table; identity $H(X Y) = H(X, Y) - H(Y)$.	Entropy + conditional/joint entropy; MI as $H(X) - H(X Y)$.
1 (b)	MI identity $I(X; X+Y) - I(Y; X+Y) = H(X) - H(Y)$ via conditional MI manipulation.	Symmetry and chain properties of (conditional) MI.
2	Lemma $I(X; Y; Z) = I(X; Y) - I(X; Y Z)$; corollary $I(X; Y) = I(X; Y Z) + I(Y; Z) - I(Y; Z X)$.	Three-variable MI: chain rule and inclusion-exclusion structure.
3	$R_{\text{emp}} = -\frac{1}{n} \sum_i [(1-\beta) \log \pi_{y^{(i)}} + \frac{\beta}{g} \sum_k \log \pi_k]$ + vectorised numpy loss.	Cross-entropy as a KL objective; entropy of target is a θ -free constant; label smoothing as a calibration tool.